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Fall 2024
UCLA

Linear Algebra, Math 115A

Exercise Sheet 7 / for Final practice

You will have 3 hours for the final and be allowed to bring one sheet (handwritten, double sided, standard letter/A4 size) of notes. The final will have 6 Exercises. The final covers all material we've covered in class, with a slight bias towards the recent. For Homework this week, turn in Ex. 7 and 8; the deadline is extended to Friday.

Exercise 1
Consider the matrix

$$A = \begin{pmatrix} 1 & 2 & 1 & -1 \\ 0 & 1 & -1 & -1 \\ 1 & 0 & 0 & 2 \\ 1 & 1 & 1 & 0 \end{pmatrix}$$

(a) Determine $\det(A)$.

(b) Determine A^{-1}.

Exercise 2 Let V be an inner product space over $\mathbb{R}$, and $f : V \to V$ a linear map such that $\langle f(v), w\rangle = \langle v, f(w)\rangle$ for all $v, w \in V$.

Show that if $v_1, v_2 \in V$ are eigenvectors of f with different eigenvalues, then $\langle v_1, v_2\rangle = 0$.

Exercise 3
Consider the $\mathbb{R}$-vector space $\mathbb{R}^{2\times 2}$ of 2×2- matrices, and the map

$$\langle A, B\rangle = \operatorname{tr}(AB^T),$$

where $\operatorname{tr}(\begin{pmatrix} a & b \\ c & d \end{pmatrix}) = a + d$ and $\begin{pmatrix} a & b \\ c & d \end{pmatrix}^T = \begin{pmatrix} a & c \\ b & d \end{pmatrix}$ denotes the transpose.

(a) Show that $\langle \cdot, \cdot\rangle$ defines an inner product on $\mathbb{R}^{2\times 2}$.

(b) Use the Gram-Schmidt process to turn $\left(\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ -1 & 1 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}\right)$ into an ONB with respect to the above inner product.

Exercise 4
Let $k \geq 1 \in \mathbb{N}$ and let $V = \mathbb{R}[T]_{\leq k}$ be the space of polynomials over $\mathbb{R}$ of degree $\leq k$.

(a) Show that $\langle f, g\rangle = \int_{-1}^{1} f(x)g(x)dx$ is an inner product on V.

(b) Let $k = 3$. Use Gram-Schmidt orthonormalization to produce an orthonormal basis with respect to the above inner product starting with the set $(T^3, T^2, T, 1)$.

Exercise 5
Consider the linear map

$$f : \mathbb{R}^3 \to \mathbb{R}^3 : \begin{pmatrix} x \\ y \\ z \end{pmatrix} \mapsto \begin{pmatrix} 2x - 2y + z \\ x + 2z \\ 2y + 3z \end{pmatrix}.$$

(a) Determine bases for $\ker(f)$ and $\operatorname{im}(f)$.

(b) Is f injective and/or surjective?

Exercise 6 Let V be a K-vector space and let $f : V \to V$ be a linear map such that $f \circ f = id_V$, i.e. such that $f(f(v)) = v$ for all $v \in V$.

(a) Show that for every $v \in V$, $v + f(v) \in E_1(f)$, and $v - f(v) \in E_{-1}(f)$.

(b) Show that $V = E_1(f) + E_{-1}(f)$, i.e. that every vector in V can be written as the sum of a vector in $E_1(f)$ and a vector in $E_{-1}(f)$.

Exercise 7 Let $v = \begin{pmatrix} v_1 \\ v_2 \\ v_3 \end{pmatrix}$ and $w = \begin{pmatrix} w_1 \\ w_2 \\ w_3 \end{pmatrix} \in \mathbb{R}^3$ be fixed.

(a) Show that the map $f : \mathbb{R}^3 \to \mathbb{R}$, $f(u) = \det(u, v, w)$ is linear.

(b) Write the map f in coordinates using the standard bases for $\mathbb{R}^3$ and $\mathbb{R}$.

(c) Using (b), find a vector $x \in \mathbb{R}^3$ such that $f(u) = \langle u, x\rangle$ for all $u \in \mathbb{R}^3$, where $\langle \cdot, \cdot \rangle$ is the standard inner product.

Remark: The vector x is then also called the 'cross product' of v and W and written $v \times w$.

Exercise 8
Let $V = \mathbb{R}[T]_{\leq 2}$ and consider $f : V \to V$ given by $f(p) = p(0) + Tp(1) + T^2p(2)$.

(a) Find $[f]_\beta^\beta$, where $\beta = (T^2, T, 1)$.

(b) Find the eigenvalues of f and their multiplicities and determine whether f is diagonalizable. If it is, determine a basis of eigenvectors.